

B.Tech. Degree IV Semester Examination April 2014**ME 1405 HYDRAULIC MACHINERY**
(2012 Scheme)

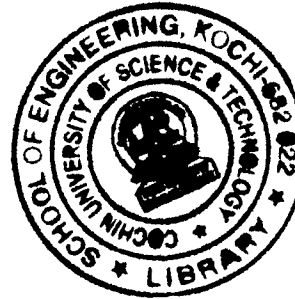
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) State Buckingham's pi-theorem.
- (b) A blade turns the jet of diameter 3cm at a velocity of 20m/s by 60°. Determine the force exerted by the blade on the fluid.
- (c) Name and briefly describe the differences between the two basic types of dynamic turbine.
- (d) What is a draft tube and what is its purpose? Describe what would happen if turbomachinery designers did not pay attention to the design of the draft tube.
- (e) What is cavitation in centrifugal pumps?
- (f) What are the advantages of installing air vessels in reciprocating pumps?
- (g) Explain the working of a vane pump with neat sketch.
- (h) Write short note on hydraulic ram.

**PART B**

(4 x 15 = 60)

- II. Derive on the basis of dimensional analysis suitable parameters to present the thrust developed by a propeller. Assume that the thrust P depends upon the angular velocity ω , speed of advance V , diameter D , dynamic viscosity μ , mass density ρ , elasticity of the fluid medium which can be denoted by speed of sound C in the medium. (15)
- OR**
- III. A water jet 20mm in diameter and having a velocity of 90m/s strikes series of moving blades in a wheel. The direction of the jet makes 20° with the direction of movement of the blade. The blade angle at inlet is 35°. If the jet should enter the blade without striking, what should be the blade velocity? If the outlet angle of the blade is 30°, determine the force on the blade. Assume that there is no friction involved in the flow over the blade. (15)
- IV. A pelton turbine is required to develop 9000 KW when working under a head of 300m the impeller may rotate at 500rpm. Assuming a jet ratio of 10 and an overall efficiency of 85%, calculate. (15)
 - (i) Quantity of water required
 - (ii) Diameter of the wheel
 - (iii) Number of jets
 - (iv) Number and size of the bucket vanes on the runner

OR

(P.T.O.)

- V. (a) Derive the expression for specific speed of a turbine. (7)
 (b) A reaction turbine working under a head of 100m and at a speed of 700rpm has an overall efficiency of 80%. If the specific speed is 175, calculate the discharge. What would be required if the head is reduced to 80m? (8)
- VI. (a) A centrifugal pump is discharged $118\text{m}^3/\text{sec}$ at a speed of 1450rpm against a head of 25m. The impeller diameter is 25cm, its width at outlet is 5cm and the manometric efficiency is 0.75. Determine the vane angle at the outer periphery of the impeller. (8)
 (b) Write a short note on multistage centrifugal pump. (7)
- OR**
- VII. (a) Explain the function of air vessels fitted to a reciprocating pump. (9)
 (b) A single acting reciprocating pump has a plunger of diameter 50cm and a stroke of 80cm. If the speed of the pump is 60rpm and coefficient of discharge is 0.97, determine the actual discharge and the percentage slip of the pump. (6)
- VIII. Explain construction and working of hydraulic torque convertor with neat sketch, (15)
 style="text-align: center;">**OR**
- IX. Write short notes on: (15)
 (i) Hydraulic accumulator
 (ii) Hydraulic intensifier
